2025-26 Part III/MASt Astrophysics Research Project: Interim Report Coversheet

For completion by the student:					
Project Title:	Euclidifying Gaia				
Student Name:	Siwei Zhao	**Signature:**	Siwei Zhao	**Date:**	1 Dec 2025
Student Declaration:	I declare that this interim report represents work undertaken as part of the NST Part III Astrophysics/MASt Astrophysics Research Project. It is the result of my own work and, includes nothing which was performed in collaboration. Any content generated by artificial intelligence has been acknowledged accordingly. No part of the report has been submitted for any degree, diploma, or any other qualification at any other university. I also declare that an electronic PDF file containing this cover sheet and work has been uploaded to the Part III/MASt Astrophysics Moodle site on this date.				

For completion by the Supervisor(s):					
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Supervisor 1 Declaration:	I declare that I have reviewed this interim report prior to submission by the student.				
Supervisor 2 Name:		**Signature:**		**Date:**	
Supervisor 2 Declaration:	I declare that I have reviewed this interim report prior to submission by the student.				

1 Progress Summary

This initial phase established the foundational infrastructure and validation necessary for subsequent machine learning analysis. Two essential components were completed: establishing the Euclid-Gaia cross-matching methodology, and validating which Euclid morphological classifiers can serve as reliable reference standards for training.

Euclid-Gaia cross-matching: The Euclid Q1 catalogue includes pre-existing Gaia DR3 associations stored in `Gaia_id`, computed via sophisticated epoch-propagated position matching, with maximum allowed distance of 0.3 arcsec.[1] ERO sources were matched using simpler position-based matching with `m.v_alpha_j2000` and `m.v_delta_j2000`. The optimal matching radius of 0.1 arcsec was confirmed through statistical analysis.

Euclid classifier validation: Using Euclid ERO data, the `class_star` and `spread_model` classifiers were validated across $10 < G < 28$ mag. `spread_model` exhibits systematic magnitude-dependent bias rendering it unreliable, while `class_star` provides robust classification across all magnitude ranges. This establishes `class_star` as the reference standard for subsequent Gaia source classification.

2 Preliminary Findings

2.1 Magnitude Selection for Cross-Survey Comparison

To enable consistent comparison across Euclid ERO and Q1 catalogues, magnitude measurements were selected based on analogous photometric roles. The selected magnitudes represent each survey's primary broad-band "total" flux measurement:

Euclid ERO: `v_mag_auto & v_flux_auto` from `merge_cat`
– Kron-like elliptical aperture magnitude in the VIS detection image, measured directly with SExtractor's automatic aperture photometry routine,[2] derived from Kron's first moment algorithm.

Euclid Q1: `flux_detection_total` from `merge_cat`
– Kron aperture photometry on the VIS/NIR stacked detection image. This quantity is explicitly defined as equivalent to `flux_auto` in SExtractor,[3] and it serves as the primary total flux measurement used by the PHZ classification pipeline.[4]

Gaia DR3: `phot_g_mean_mag & phot_g_mean_flux` from `gaia_source`
– PSF-fitted mean magnitude in Gaia's broad optical G band (330 – 1050 nm), serving as the primary photometric reference for all Gaia sources.

The Q1 and ERO magnitudes are directly comparable: both represent Kron "total" flux in their respective detection bands. This consistency enables cross-validation of morphological classifiers between Q1 and ERO datasets. Gaia's G-band magnitude, while produced with PSF fitting rather than aperture photometry and spanning a broader wavelength range than Euclid VIS, serves as a practical reference for photometric consistency checks given its role as Gaia's primary broad-band measurement.

2.2 Euclid ERO Classifiers: `class_star` vs `spread_model`

Sanity check was performed for both Euclid ERO classifiers across $12 < G < 28$ mag. The `class_star` classifier shows the expected monotonic behavior: point-like sources dominate at bright magnitudes, with extended fraction increasing smoothly toward fainter magnitudes. In contrast, `spread_model` exhibits a characteristic bumpy structure caused by magnitude-dependent

systematic effects: it uses an error-weighted threshold that depends on its own the uncertainty `spreaderr_model`, where the threshold for star-galaxy separation is calculated by

$$\sqrt{\epsilon^2 + (\kappa \times \texttt{spreaderr_model})^2}$$

where ϵ and κ are constants. Since `spreaderr_model` is inversely related to SNR, this creates magnitude-dependent bias:

- **Bright regime** ($12 < G < 18$ mag): High SNR measurements produce low `spreaderr_model`, thus tight astrometric threshold for star-galaxy separation. Genuinely point-like sources with small positive `spread_model` values exceed this tight threshold and are misclassified as extended, producing the first bump.
- **Intermediate regime** ($18 < G < 23$ mag): Temporarily reasonable classification where `spread_model` performs adequately as SNR-dependent threshold effects balance out, but this narrow range maybe insufficient to general use.
- **Faint regime** ($G > 23$ mag): Low SNR produces high `spreaderr_model` values, loosens the threshold. Extended sources with noise-inflated `spread_model` values fall below this loose threshold and are misclassified as point-like, producing the second bump.

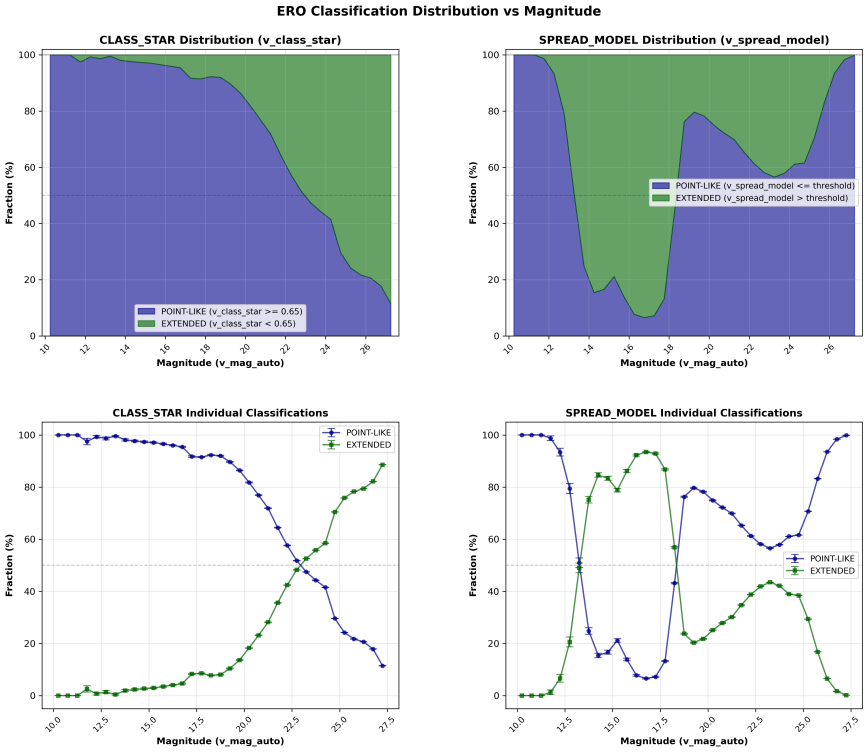


Figure 1: ERO classification distribution vs magnitude. Top: Stacked area plots showing point-like (blue) and extended (green) fractions. Bottom: Individual classification fractions with binomial uncertainties.

This error-weighted threshold approach, while intended to maintain consistent classification reliability across magnitude ranges, instead introduces systematic bias by making the classification criterion itself magnitude-dependent rather than morphology-dependent, rendering it unreliable across magnitude ranges. Thus we determined that classifier `class_star` is preferred over `spread_model` for extended source identification.

2.3 Q1 Photometric Classification

Euclid Q1 includes PHZ pipeline classifications derived from photometric redshift analysis using multi-band photometry. Classification distribution was analysed as a function of magnitude, where AB magnitudes are derived from fluxes using:

$$m_{\mathrm{AB}} = -2.5 \times \log_{10}(\mathrm{flux}_{\mu\mathrm{Jy}}) + 23.9$$

where 23.9 is the magnitude zero point.[3]

Figure 2 shows classification trends across magnitude. At mag $\gtrsim 20$, PHZ classifications exhibit physically reasonable behavior: GALAXY classification (I2, green) rises steadily toward fainter magnitudes, approaching ∼99% at the survey depth limit, consistent with the expected dominance of extended extragalactic sources at faint flux levels.

However, at mag < 20, the combined QSO+GLOBULAR_CLUSTER+OTHER category dominates at ∼85–100%, indicating classification uncertainty at the bright end, likely due to small number statistics and potential saturation effects. Notably, many bright point sources may fall into this "OTHER" category rather than being classified as stars. For this analysis, the key observation is that the bright regime is dominated by point-like sources as opposed to extended galaxies, which is physically expected. The lack of detailed PHZ sub-classification at bright magnitudes does not impact subsequent extended source identification, thus a magnitude cut at mag ∼20 may be appropriate for reliable extended classifications.

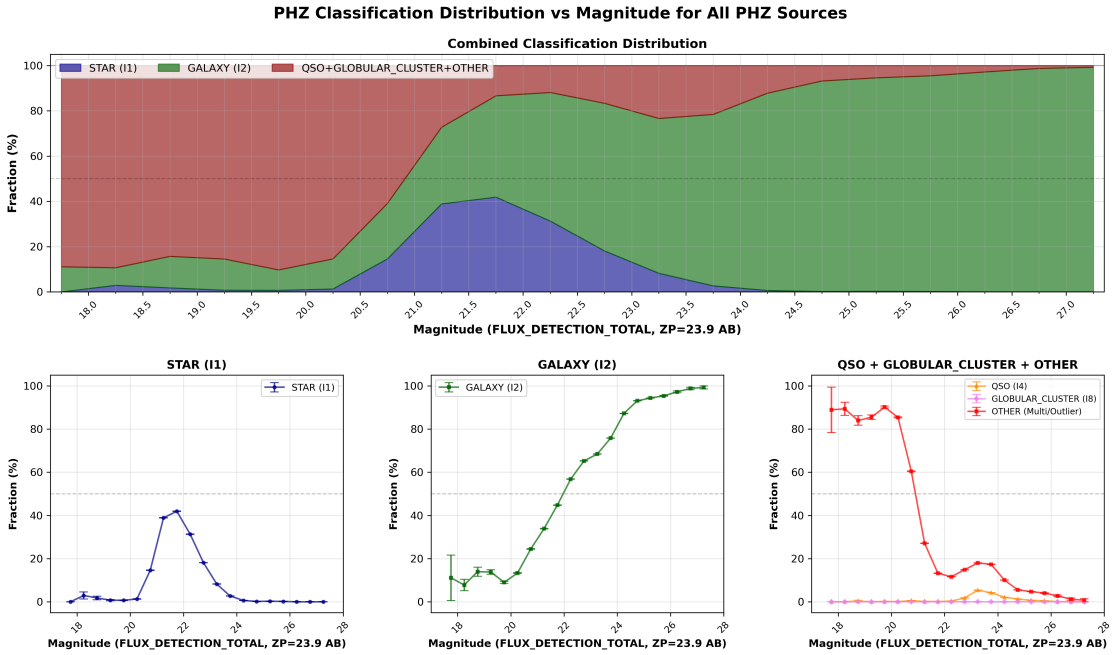


Figure 2: PHZ classification distribution vs magnitude. Top: Stacked area plot showing STAR (I1, blue), GALAXY (I2, green), and combined QSO+GLOBULAR_CLUSTER+OTHER (red) fractions. Bottom: Individual classification fractions with binomial uncertainties.

3 Cross-Matching Methodology

3.1 Q1-Gaia Matching Radius Validation

Angular separation distributions between Q1 and Gaia DR3 matched sources were analysed to validate the pre-computed `Gaia_id` associations.

Rayleigh distribution fitting was performed to statistically characterize this distribution. The model was fitted to the core distribution of 90th percentile to avoid contamination from spurious associations in the tail. Figure 3 shows the Rayleigh fit yields a scale parameter $\sigma = 0.091$ arcsec, thus the adopted 0.3 arcsec threshold corresponds to approximately 3σ.

Statistical tests show deviations from ideal Rayleigh behavior at large separations, likely reflecting known systematic astrometric effects in Gaia and Euclid catalogues such as parallax zero-point offsets, proper motion systematics, and catalogue-specific distortions. Despite imperfect fit at the tail, the Rayleigh model adequately describes the core distribution, providing statistical support for the 0.3 arcsec threshold in Q1-Gaia matching.

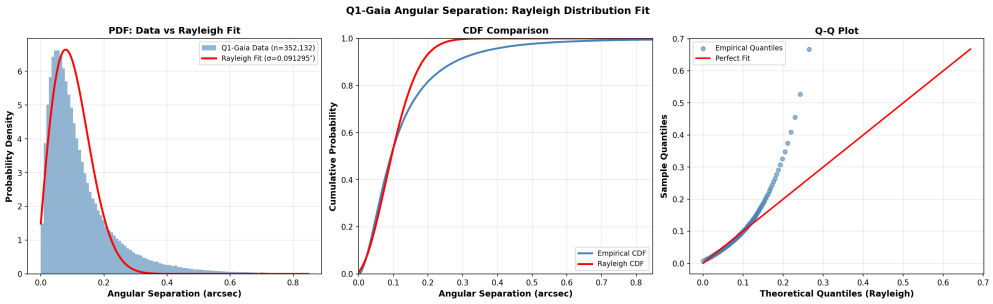


Figure 3: Rayleigh fit to Q1-Gaia core separations. Fit performed on 90th percentile. Left: PDF comparison with fitted Rayleigh. Middle: CDF comparison. Right: Q-Q plot showing good core agreement with expected tail deviation.

3.2 ERO-Gaia Matching Radius Determination

For ERO sources, position-based matching to Gaia DR3 was required in the lack of pre-computed associations. Initial testing at 0.3 arcsec, following Q1's validated threshold, revealed systematic issues: the angular separation distribution exhibits a prominent bump transition at ∼0.1 arcsec as shown in Figure 4, indicating contamination from spurious alignments beyond this radius. This bimodal structure suggests 0.3 arcsec is too loose for ERO-Gaia matching.

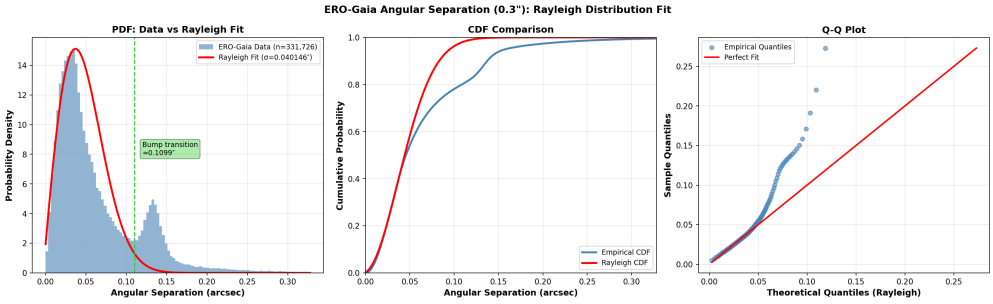


Figure 4: ERO-Gaia matching at 0.3 arcsec threshold. Left: PDF with bump transition at ∼0.1 arcsec (green dashed line). Middle: CDF comparison. Right: Q-Q plot showing unexpected jagged deviation from Rayleigh at large separations.

This contamination is absent in Q1-Gaia, implying a more sophisticated epoch-propagated matching methodology was employed for producing `Gaia_id` than just simple position-based matching.

A tighter matching radius of 0.1 arcsec was therefore adopted for ERO-Gaia crossmatching. Rayleigh fitting to this restricted sample yields $\sigma = 0.033$ arcsec, significantly tighter than Q1-Gaia $\sigma = 0.091$ arcsec. The distribution shows no bimodal structure, confirming that a 0.1 arcsec threshold effectively excludes spurious associations while retaining genuine matches.

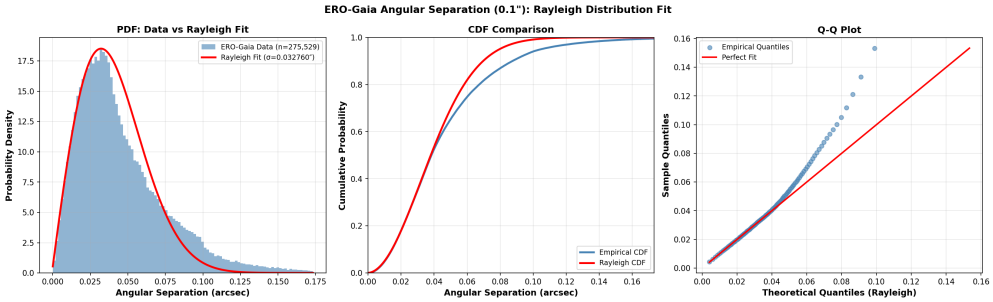


Figure 5: ERO-Gaia matching at 0.1 arcsec threshold. Rayleigh fit to 90th percentile yields $\sigma = 0.033$ arcsec and a clean single-peaked distribution.

3.3 Photometric Consistency Check

To validate the statistically-determined matching radius, photometric consistency was examined across multiple radii. Figure 6 shows magnitude differences `[Euclid Mag - Gaia Mag]` as a function of angular separation for both Q1-Gaia and ERO-Gaia crossmatches.

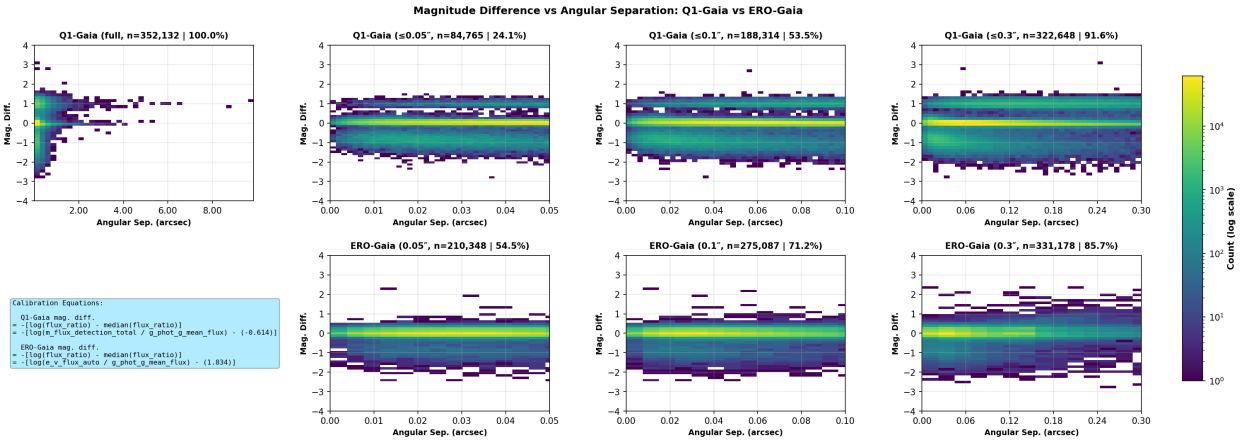


Figure 6: Photometric consistency check at multiple radii. Magnitude difference vs angular separation for Q1-Gaia (top row) and ERO-Gaia (bottom row) at 0.05, 0.1, and 0.3 arcsec. Calibration equations shown in lower left.

For ERO-Gaia, the 0.1 arcsec radius captures 71.2% of sources while maintaining photometric consistency with no systematic offset. The tighter 0.05 arcsec radius achieves higher purity but sacrifices completeness at 54.5%. The looser 0.3 arcsec radius achieves 85.7% completeness but includes significant contamination from spurious alignments at large separations, as identified in the angular separation distribution analysis.

For Q1-Gaia, the 0.3 arcsec threshold captures 91.6% of sources. However, there is an apparent bimodality in Q1-Gaia magnitude differences, reflecting the use of `flux_detection_total`, which

gives VIS/NIR stack band source total flux, when matching to `phot_g_mean_mag`, a calibrated mean G-band flux. This is accounted for in PHZ processing through per-band zero-point multiplicative corrections (ZPC).[4] ERO uses only VIS-band photometry `v_mag_auto`, avoiding this multi-band complexity and showing no bimodality.

4 Future Work & Timeline

4.1 Upcoming Analysis

Photometric Systematics: Characterize magnitude differences Δmag (Euclid − Gaia) as functions of magnitude, color (BP−RP), and morphological class to identify systematic photometric offsets. Investigate extended source distribution in color-magnitude space.

Machine Learning: Identify optimal Gaia parameters for machine learning, train XGBoost classifier using Euclid labels, and perform feature importance analysis to identify key diagnostic parameters.

Extended Source Characterization: Apply clustering methods (UMAP/self-organizing maps) for morphological sub-classification. Extract physical properties (size, surface brightness) from Euclid images and cross-validate classifications.

Full Catalogue Application: Deploy validated classifier to Gaia DR3, generate extended source catalogue with morphological types, compare population demographics with Euclid overlap regions, analyse spatial distribution, clustering properties, and physical interpretations by classifications.

4.2 Project Goals

Primary objective: Develop and validate machine learning pipeline for identifying and classifying extended sources in Gaia DR3 using Euclid training data.

Deliverables: (1) Validated XGBoost classifier for Gaia extended source identification, (2) feature importance analysis identifying key Gaia morphological indicators, (3) morphological sub-classification scheme for extended sources, (4) Gaia extended catalogue with classifications.

If time allows: Detailed physical characterization using Euclid imaging for representative samples, investigation of environmental dependence of extended source morphologies, comparison with existing catalogues.

References

[1] Cuillandre et al. (2025). Euclid Quick Data Release (Q1): From images to multiwavelength catalogues: The Euclid MERge Processing Function. https://doi.org/10.48550/arXiv.2503.15305

[2] SExtractor User Manual. https://sextractor.readthedocs.io/en/latest/Photom.html

[3] Euclid SGS Data Product Description Document: MER Photometry Cookbook. http://st-dm.pages.euclid-sgs.uk/-/data-product-doc/-/jobs/162527/artifacts/build/merdpd/merphotometrycookbook.htmll

[4] Cuillandre et al. (2025). Euclid Quick Data Release (Q1): Photometric redshifts and physical properties of galaxies through the PHZ processing function. https://arxiv.org/pdf/2503.15306